

Practice test 14A

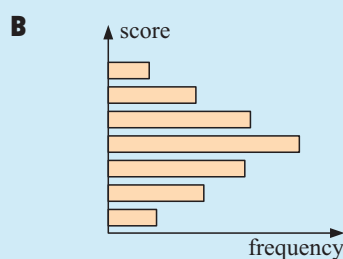
Multiple Choice

1 Which of the following is a continuous numerical variable?

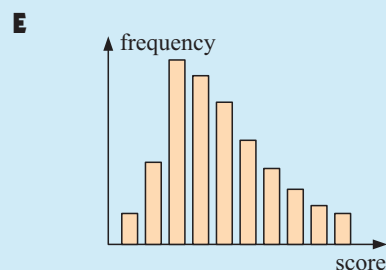
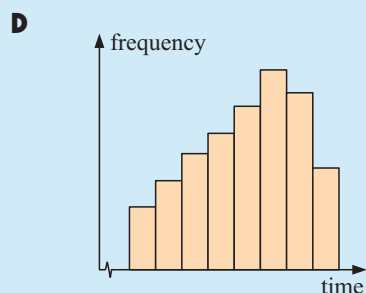
- A** the favourite movie of Year 9 students
- B** the weights of cupcakes in a bakery
- C** the marks obtained in a mathematics test
- D** the number of cars in Australian households
- E** the suburb where patrons of a store live.

2 Which of these data sets is positively skewed?

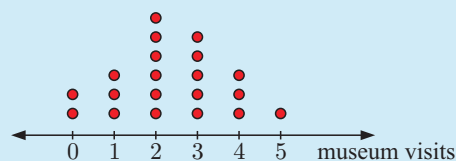
A Stem	Leaf
0	2
1	0 0 4 3
2	8 9
3	1 2
4	3 3 3 4 5 5 8
5	6



C 0, 0, 1, 1, 2, 2, 3, 3, 4, 4



Questions 3 and 4 relate to a survey of people standing in the foyer of a museum. The number of times they had previously visited the museum in the past five years is summarised in the dot plot:



3 How many people were surveyed?

- A** 5
- B** 15
- C** 20
- D** 10
- E** 25

4 What percentage of survey respondents had visited the museum at least 3 times previously in the last 5 years?

- A** 45%
- B** 55%
- C** 20%
- D** 30%
- E** 75%

Questions **5** and **6** refer to the table alongside, which shows the average lap times for a squad of long distance runners:

<i>Time t (s)</i>	<i>Frequency</i>
$56 \leq t < 58$	2
$58 \leq t < 60$	3
$60 \leq t < 62$	4
$62 \leq t < 64$	6
$64 \leq t < 66$	8
$66 \leq t < 68$	10
$68 \leq t < 70$	5
$70 \leq t < 72$	2

5 The modal class for this data is:

- A** $60 \leq t < 62$ **B** $62 \leq t < 64$ **C** $64 \leq t < 66$
D $66 \leq t < 68$ **E** $68 \leq t < 70$

6 Runners with average lap times less than 64 seconds were selected to participate in the National Championships. What percentage of runners were selected?

- A** 5% **B** $12\frac{1}{2}\%$ **C** $22\frac{1}{2}\%$ **D** 15% **E** $37\frac{1}{2}\%$

7 The scores 12, 14, x , 8, 9, 11, 6, and 8 have a mean of 10. The value of x is:

- A** 10 **B** 11 **C** 12 **D** 13 **E** 14

Questions **8** and **9** refer to the following scenario:

Mark worked overtime every work day for 10 days. The number of minutes of overtime he worked each day were:

45 15 60 30 40 55 10 25 45 35

8 The interquartile range for this data set is:

- A** 10 minutes **B** 20 minutes **C** 30 minutes **D** 40 minutes **E** 50 minutes

9 The median of this data set is:

- A** 10 minutes **B** 20 minutes **C** 30 minutes **D** 40 minutes **E** 50 minutes

10 A dance school wishes to find out the most appropriate date for its end of term party. Which of the following methods would most likely produce an unbiased result?

- A** a questionnaire is sent to 50 randomly selected students and instructors
B the instructors are all asked for their preferred date
C the students who attend Saturday morning classes are asked for their preferred date
D the Bollywood dance students are asked for their preferred date
E the principal instructor selects a date at random.